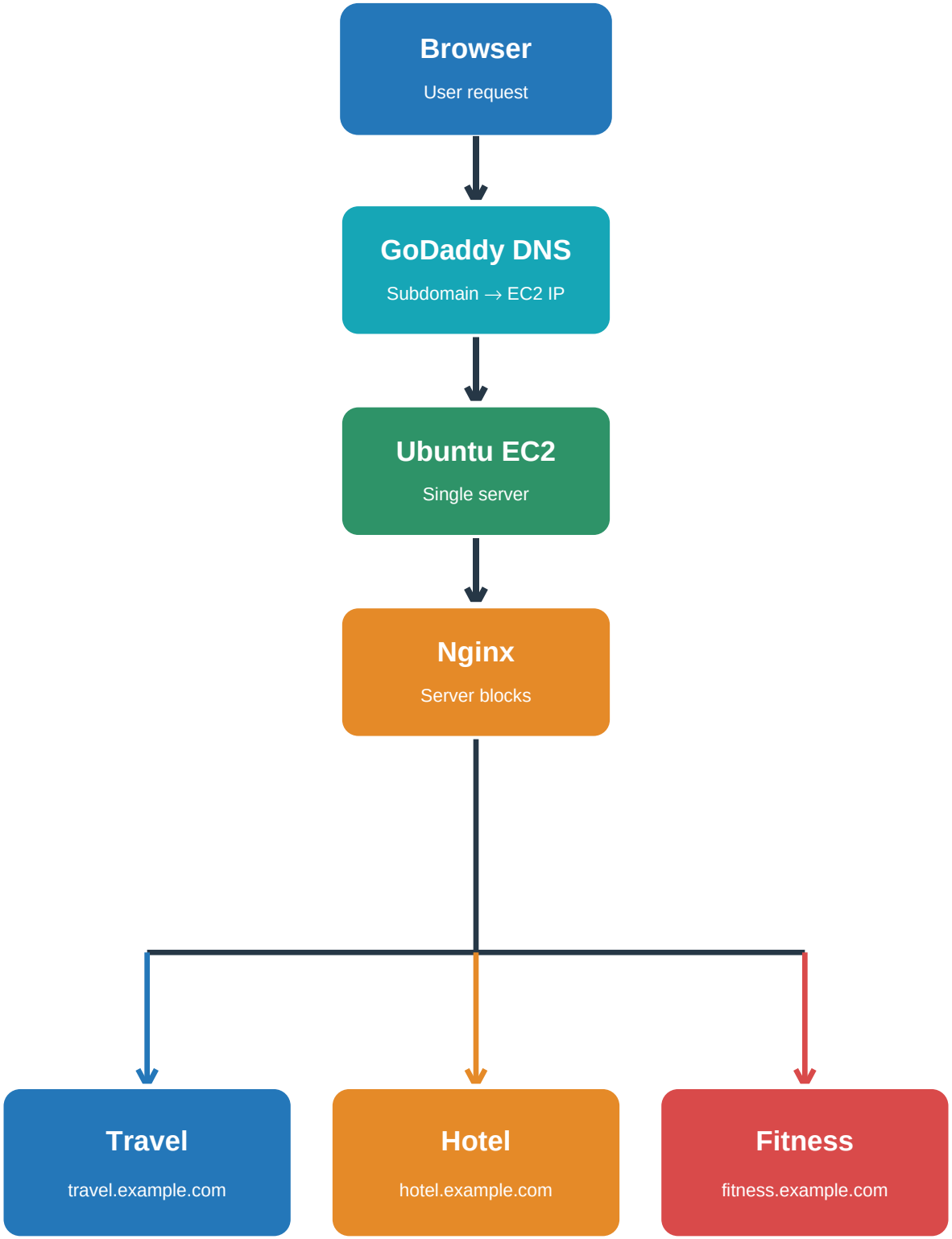
Ubuntu Nginx Multi-Site Deployment

One Ubuntu EC2 Server • GoDaddy DNS • Nginx • Three Static Websites

Deployment Architecture

Subdomain request → DNS resolution → EC2 → Nginx server block → matching website



Nginx routes each hostname to its matching website root

Travel • Hotel • Fitness are hosted independently on the same Ubuntu EC2 server.

Project Description

Ubuntu Nginx Multi-Site Deployment

This project demonstrates how three independent static websites can be hosted on a single Ubuntu EC2 server using Nginx. GoDaddy DNS is used for the subdomains, while Nginx server blocks route each hostname to the correct website directory. The websites are Travel, Hotel and Fitness, and the deployment also includes basic shell scripting and website verification.

Project Objectives

- Host three static websites on one Ubuntu EC2 instance.
- Configure Nginx server blocks for separate websites.
- Use GoDaddy DNS to connect subdomains to the EC2 public IP.

Websites Created

- Travel website
- Hotel website
- Fitness website

Technologies Used

- AWS EC2
- Ubuntu Linux
- Nginx
- GoDaddy DNS
- Shell Scripting
- HTML
- Git / GitHub

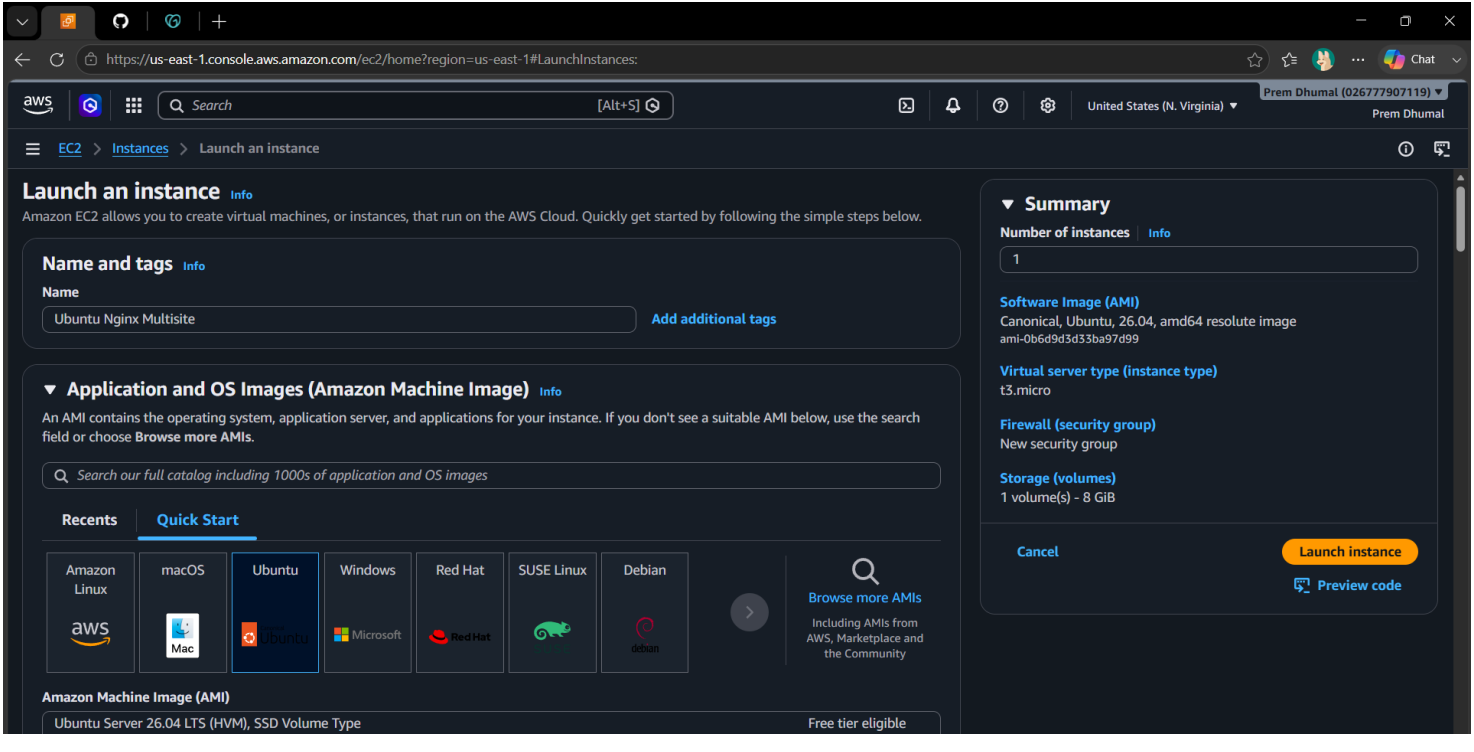
Important Paths

- /var/www/html/travel/
- /var/www/html/hotel/
- /var/www/html/fitness/
- /etc/nginx/sites-enabled/

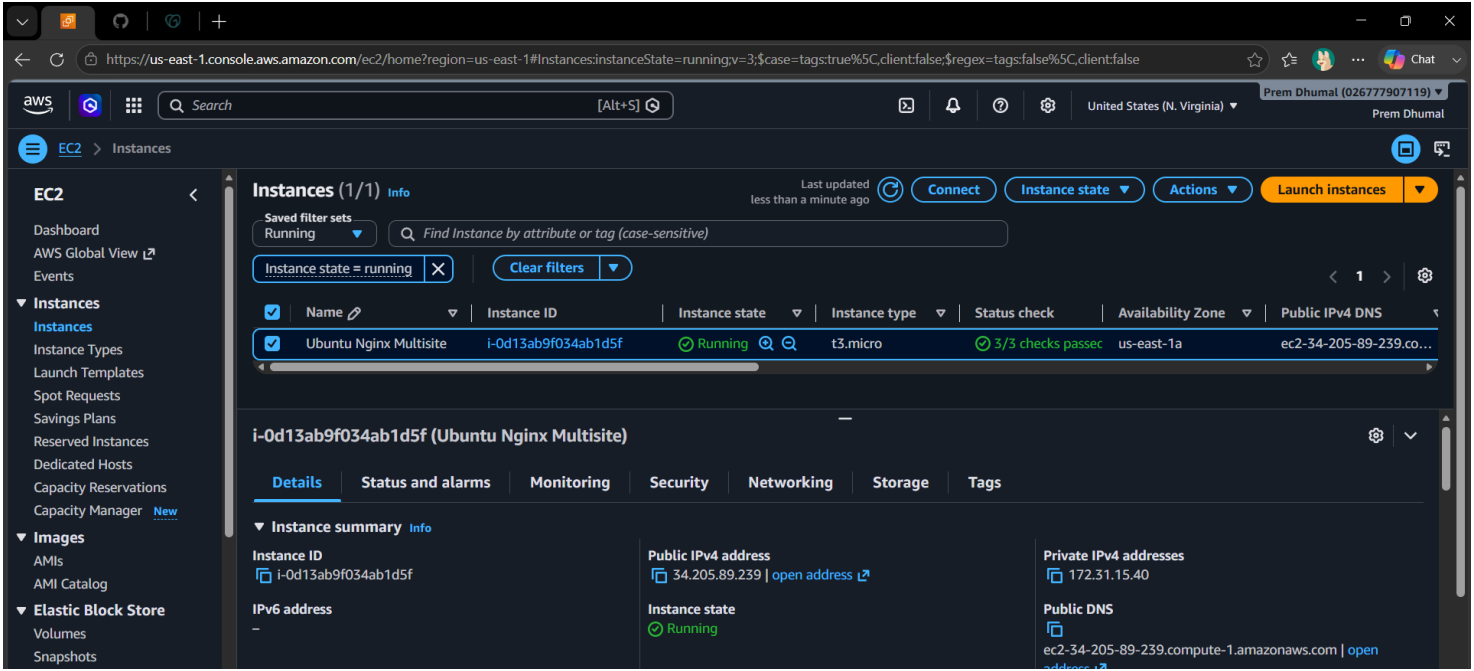
Project Result

- All three websites are kept in separate directories.
- Each subdomain is handled by its own Nginx server block.
- One Ubuntu EC2 server is used to host the complete setup.

Step 1 — Launch the Ubuntu EC2 Instance



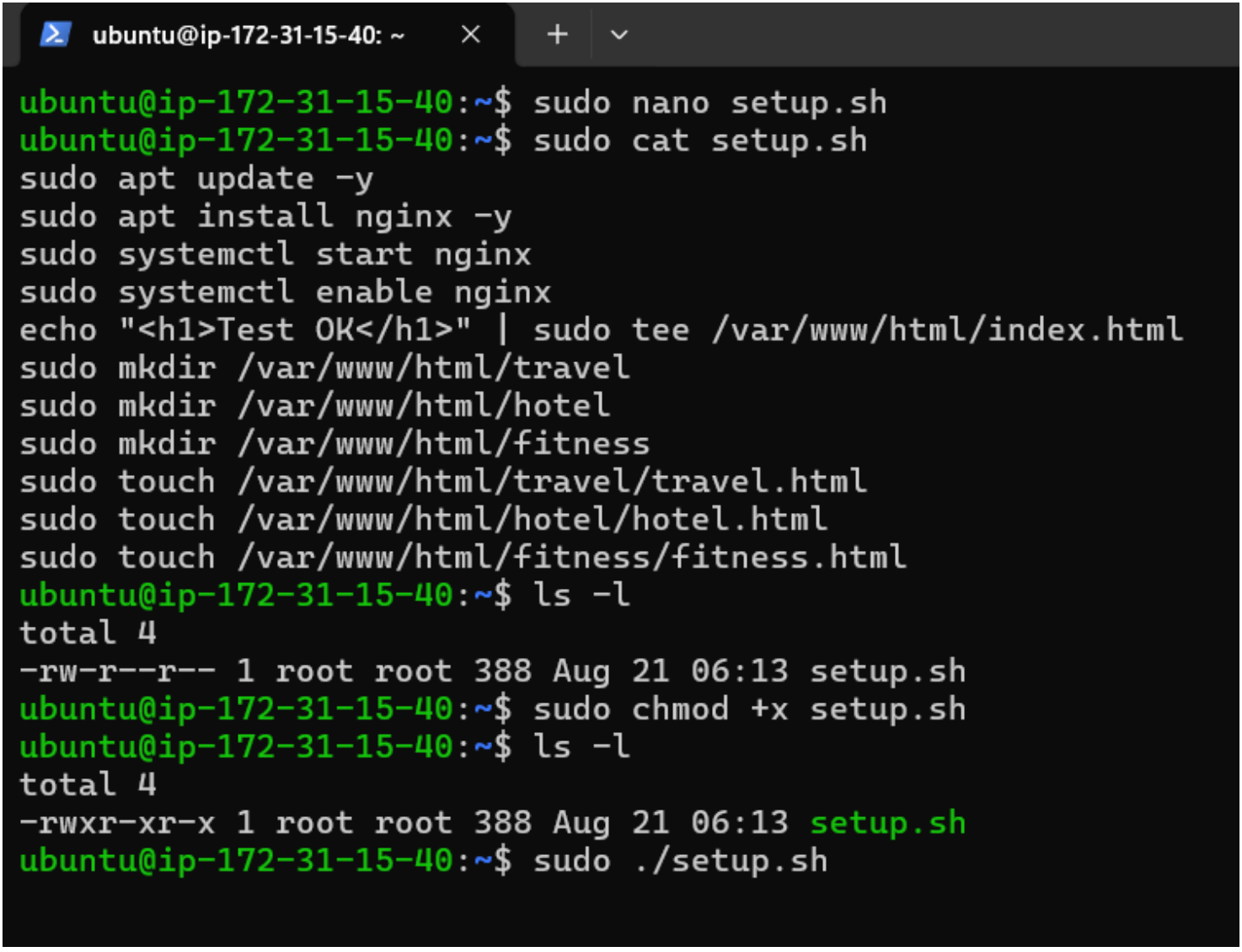
Step 2 — Verify the EC2 Instance is Running



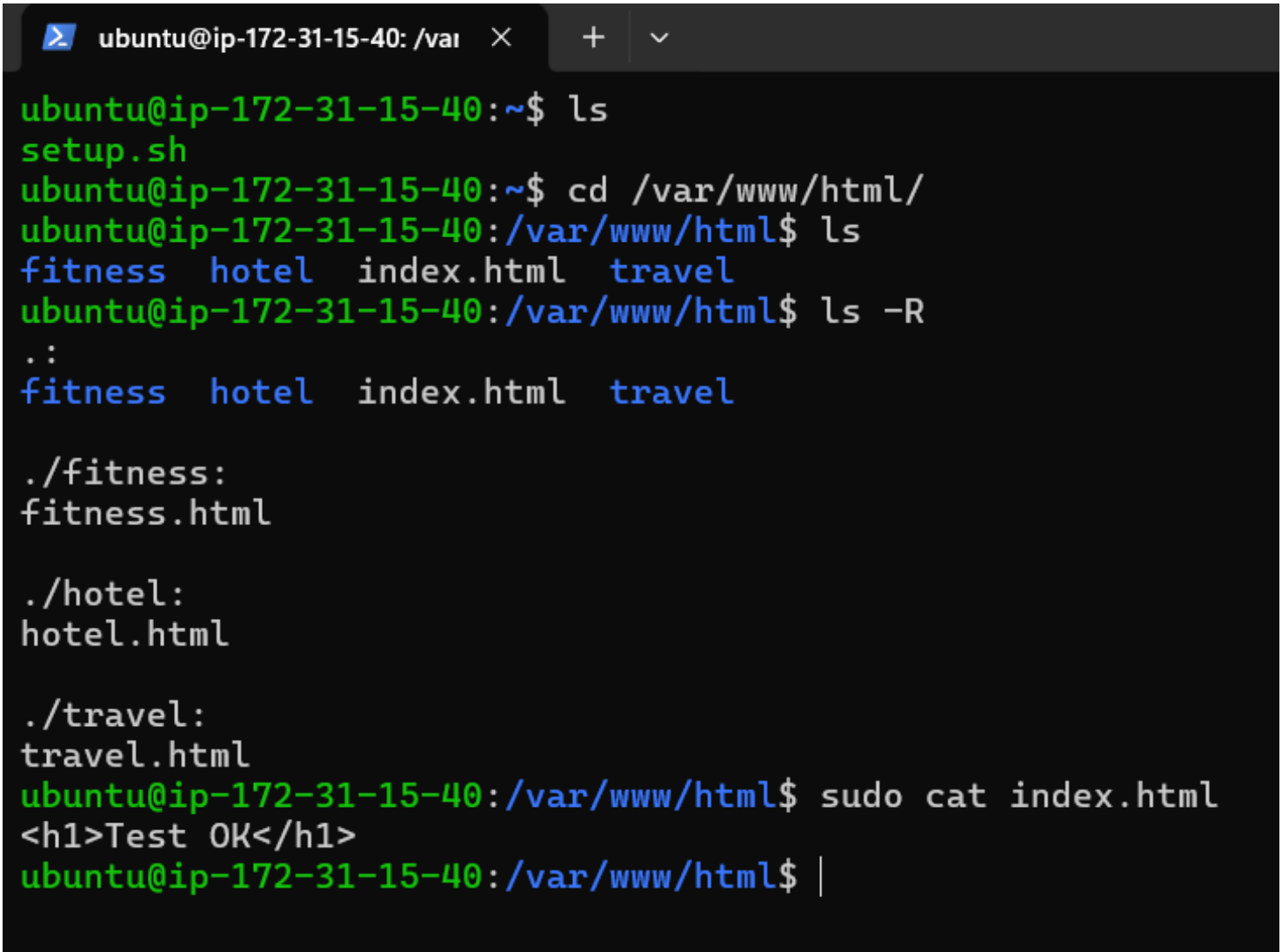
Step 3 — Connect to Ubuntu Using SSH

```
PS C:\Users\praje> ssh -i .\Downloads\ubuntu-multisite.pem ubuntu@34.205.89.239
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Aug 21 05:28:40 UTC 2026

  System load:  0.04              Temperature:           -273.1 C
  Usage of /:   30.6% of 6.61GB   Processes:             121
  Memory usage: 27%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.15.40


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Fri Aug 21 05:28:42 2026 from 152.58.33.195
ubuntu@ip-172-31-15-40:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-15-40:~$ whoami
ubuntu
```

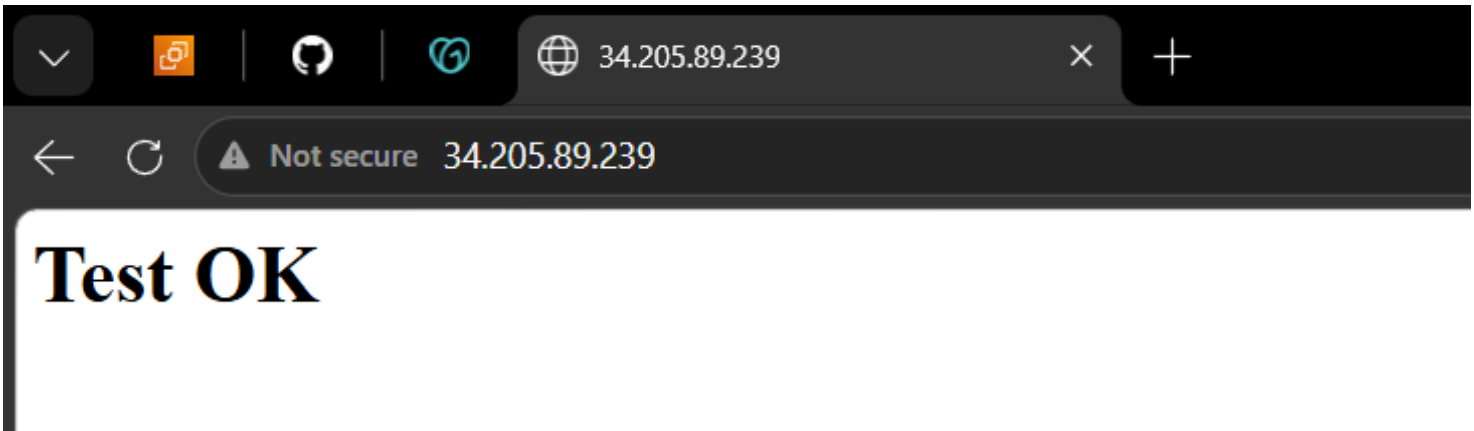
Step 4 — Create and Run the Shell Script



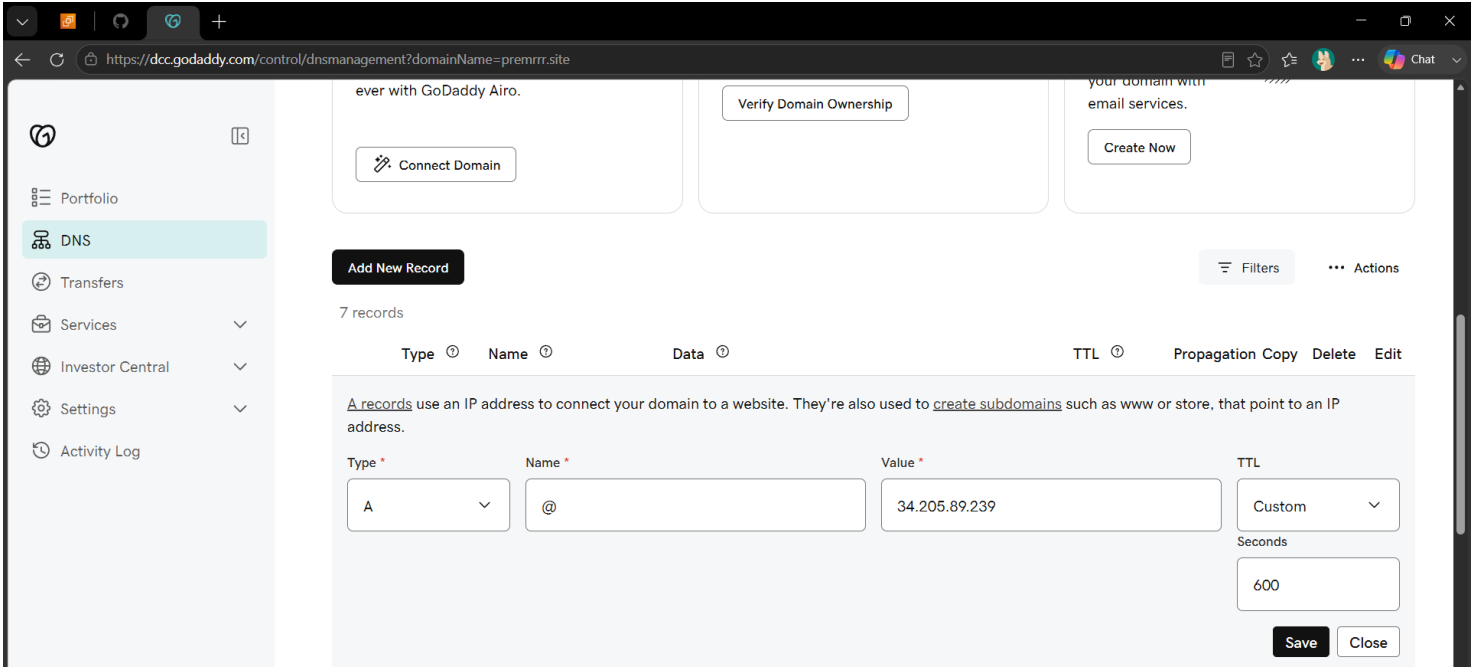
Step 5 — Verify the Website Files



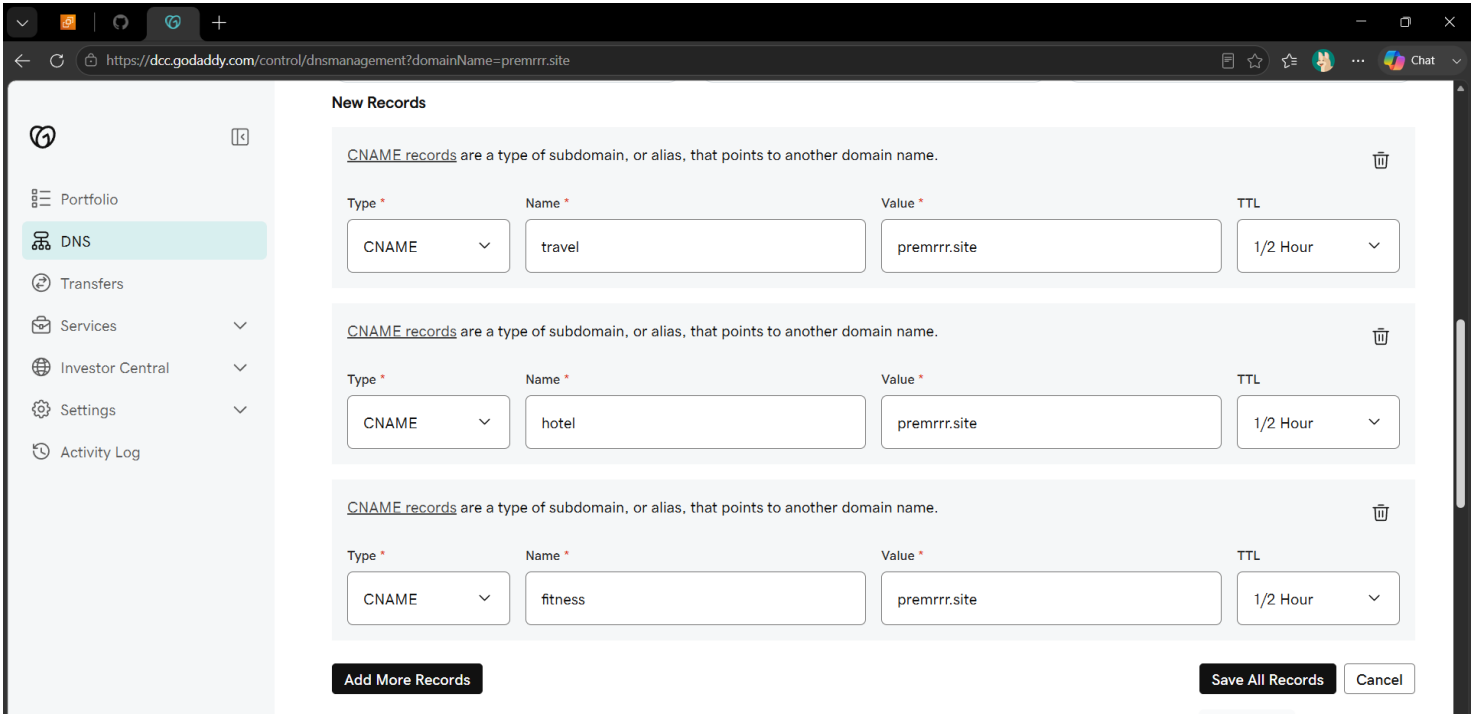
Step 6 — Confirm Shell Script Execution



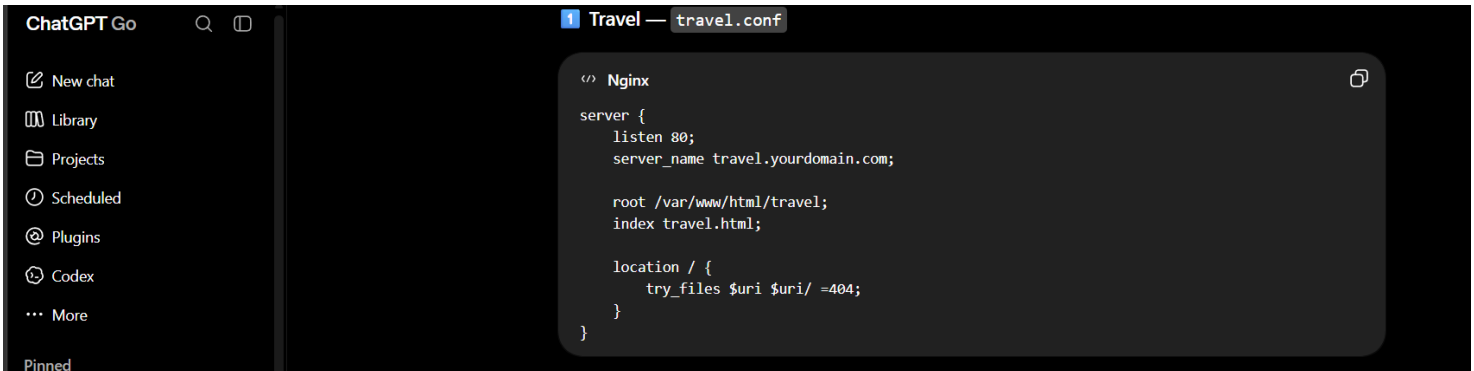
Step 7 — Add the EC2 Server IP in GoDaddy DNS



Step 8 — Create the Website Subdomains



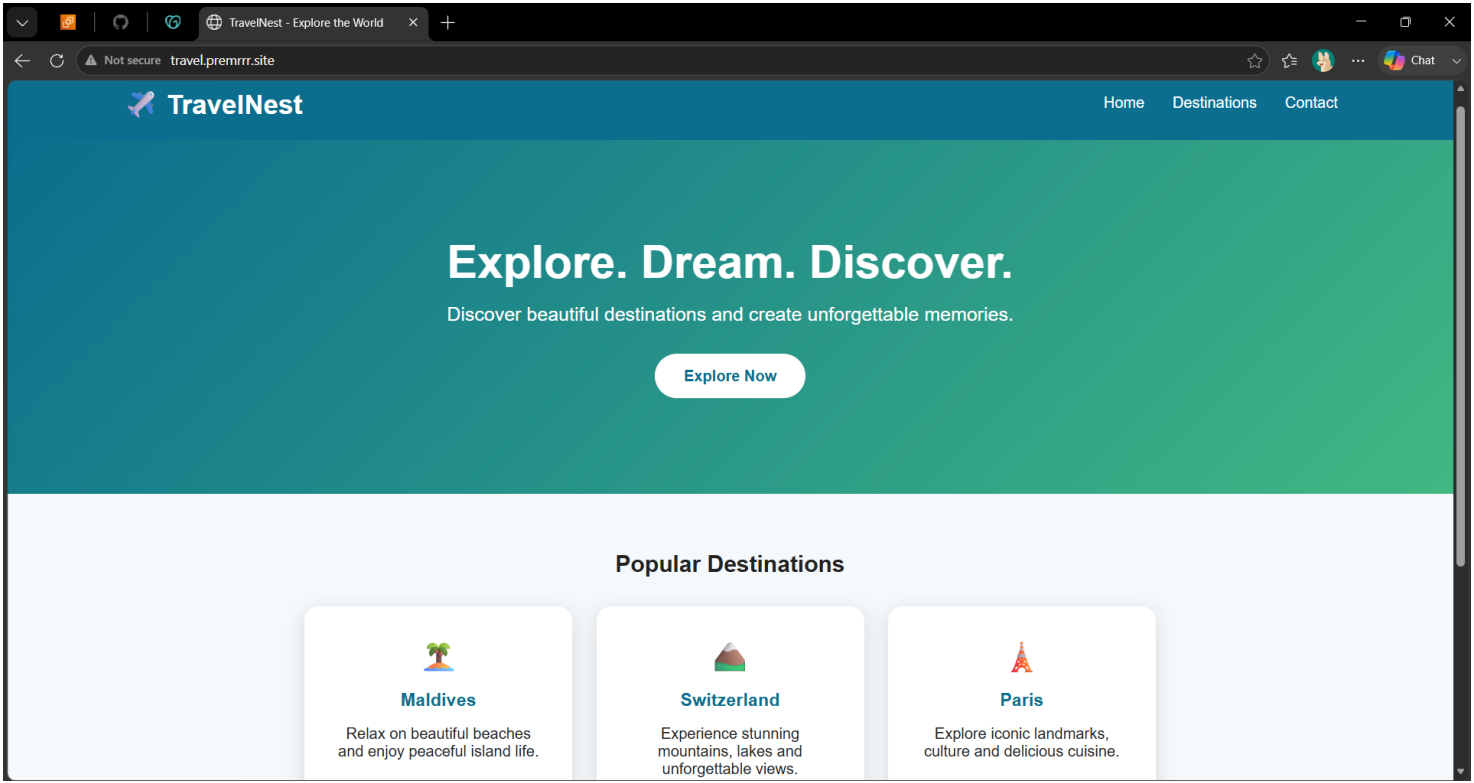
Step 9 — Prepare the Nginx Server Blocks



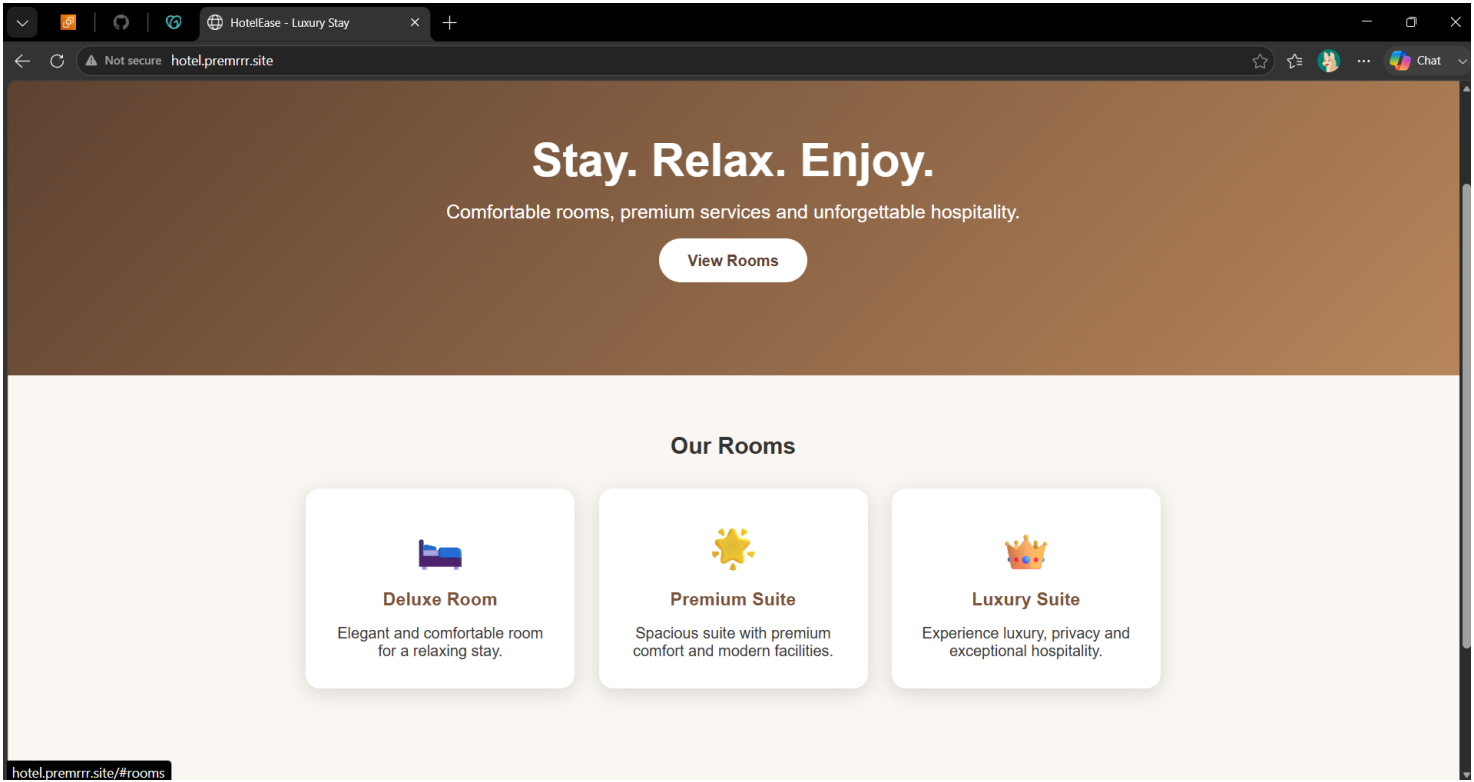
Step 10 — Create and Configure the Nginx Files

```
ubuntu@ip-172-31-15-40: /etc
ubuntu@ip-172-31-15-40:/etc/nginx/sites-enabled$ sudo nano {travel,hotel,fitness}.conf
ubuntu@ip-172-31-15-40:/etc/nginx/sites-enabled$ ls
default  fitness.conf  hotel.conf  travel.conf
ubuntu@ip-172-31-15-40:/etc/nginx/sites-enabled$ sudo cat travel.conf
server {
    listen 80;
    server_name travel.premrrr.site;

    root /var/www/html/travel/;
    index travel.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
ubuntu@ip-172-31-15-40:/etc/nginx/sites-enabled$ sudo cat hotel.conf
server {
    listen 80;
    server_name hotel.premrrr.site;

    root /var/www/html/hotel/;
    index hotel.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
ubuntu@ip-172-31-15-40:/etc/nginx/sites-enabled$ sudo cat fitness.conf
server {
    listen 80;
    server_name fitness.premrrr.site;

    root /var/www/html/fitness/;
    index fitness.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
ubuntu@ip-172-31-15-40:/etc/nginx/sites-enabled$ sudo service nginx reload
ubuntu@ip-172-31-15-40:/etc/nginx/sites-enabled$
```

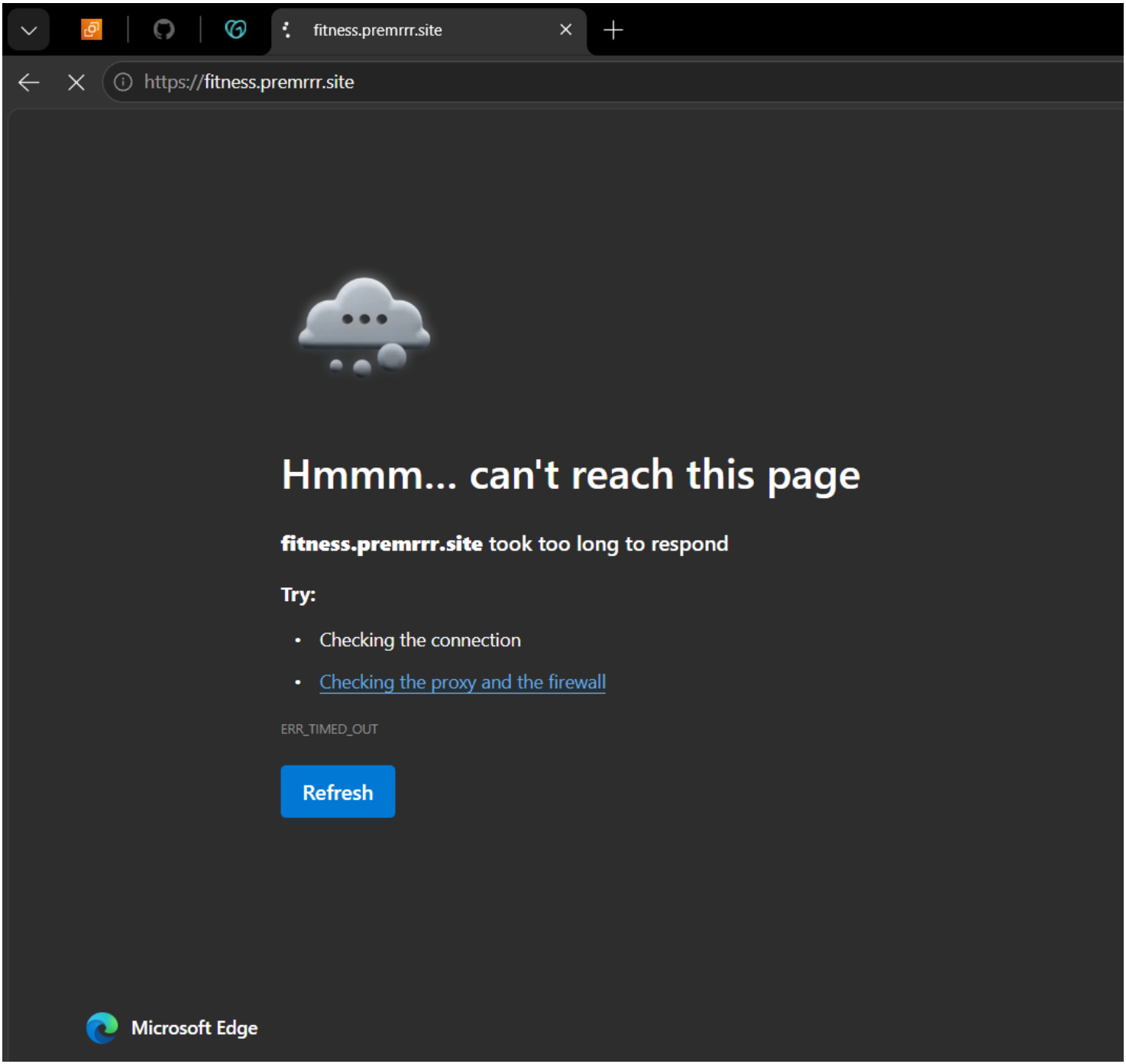
Step 11 — Verify the Travel Website



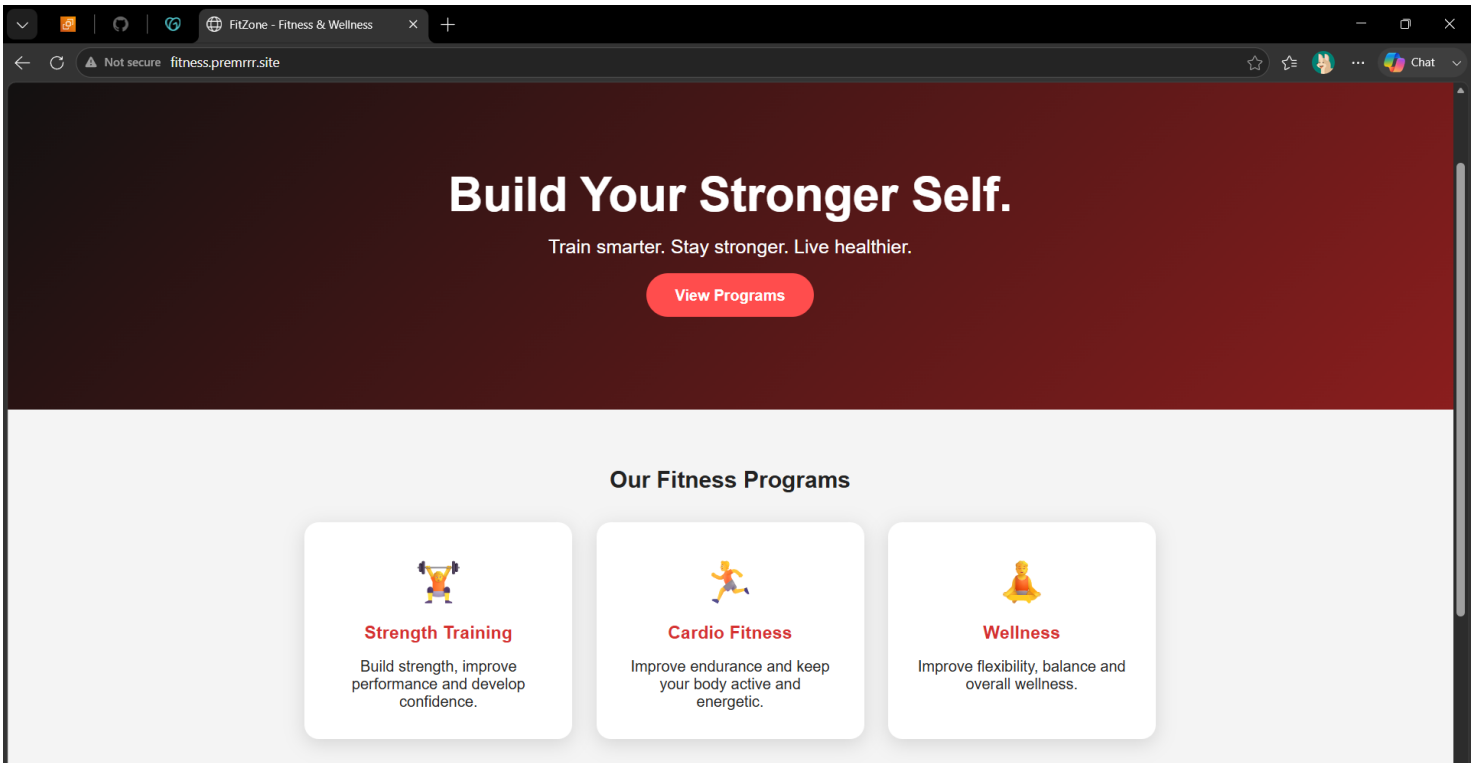
Step 12 — Verify the Hotel Website



Step 13 — Identify the HTTPS Access Issue



Step 14 — Verify the Fitness Website



Problems, Solutions & Conclusion

Problem 1 — Website access / 403

Observed: A 403 Forbidden response was encountered during website testing.

Solution: Check the Nginx server-block configuration, website root path and access permissions, then retest the configured site.

Problem 2 — HTTPS access

Observed: The project includes an HTTPS access test, but SSL/HTTPS is not configured in the deployment.

Solution: Use the configured HTTP subdomains for this project. HTTPS can be added later by configuring an SSL certificate and Nginx HTTPS settin

Problem 3 — Multiple sites on one server

Observed: A single EC2 instance must serve different websites without mixing their content.

Solution: Use separate website directories and separate Nginx server blocks, with each hostname mapped to its own root.

Conclusion

This project demonstrates a practical multi-site deployment on Ubuntu using Nginx.
One EC2 server can host Travel, Hotel and Fitness websites when DNS and server blocks are configured correctly.
The project builds hands-on understanding of EC2, Ubuntu, Nginx, DNS, subdomains, shell scripting and basic web troubleshooting.